Utilities and offsites (U-900) operating manual

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1. Purpose and scope

This manual covers the utilities that serve the process units (steam, condensate, cooling water, nitrogen, fuel and flare headers) and the water treatment area (equalization tank, slop tank, effluent). Loop details are in DOC-CN-001. Values are those of the plant's training simulator.

2. Steam and condensate

The medium-pressure steam header pressure is controlled by PIC-1217 and the steam flow to the process units by FIC-1109. The steam consumers change their demand during the day, and those changes are the main disturbance on PIC-1217 (DOC-TG-001, section 4.8).

FIC-1109 runs with its valve near fully open at high steam demand (saturation). In that state the header pressure is no longer controlled by the flow loop. Inform the shift supervisor so consumers with lower priority are reduced.

Condensate returns to the condensate drum V-1218 (25 m³, design 6 bar g / 165 °C). Its level LIC-1218 interacts with the steam network demand: when the demand rises, more condensate returns and the level rises (the SECONDARY signal of the loop on the OPC UA server).

The header pressure PIC-1008 reads the transmitter on the main header. That transmitter has a known slow drift (DOC-TG-001, section 4.6). It is calibrated every 6 months.

3. Cooling water

The cooling water pump P-1219 (1,800 m³/h rated, 630 kW) supplies the process coolers. The cooling water flow FIC-1219 runs with its valve near fully open in summer (saturation). The return temperature TIC-1220 oscillates because the consumers' own temperature loops act on the same water (interaction with consumers, DOC-TG-001, section 4.2). Do not retune TIC-1220 without checking the large consumers first.

4. Nitrogen, network and flare headers

- Nitrogen network pressure PIC-1221 is stable and needs no attention.
- Network pressure PIC-1110 and flare header pressure PIC-1212 are often kept in manual during transients (DOC-TG-001, section 4.4). Return them to automatic after the transient.

5. Water treatment

The process water collected from the units goes to the equalization tank TK-1223 (3,500 m³), whose level LIC-1223 is often kept in manual. The effluent flow to treatment FIC-1222 has its setpoint changed by the operators to follow the treatment plant capacity. The treated effluent

pH is measured by AT-1224 and controlled by AIC-1224 (setpoint pH 7.0, shown as 50 % of the analyzer span). The pH probe drifts as it ages, so the operator compares it with a lab pH every day. The legal limit for discharge is pH 6.0 to 9.0.

The slop tank TK-1225 (1,200 m³) receives recovered oil. Its level LIC-1225 is influenced by the units' load (interaction).

6. Operating limits

Variable	Low alarm	Normal	High alarm
MP steam pressure (PIC-1217)	56 bar	62 bar	68 bar
Steam flow (FIC-1109)	50 t/h	82 t/h	95 t/h
Condensate drum level (LIC-1218)	25 %	50 %	75 %
Cooling water flow (FIC-1219)	60 m³/h	81 m³/h	95 m³/h
Cooling water return temperature (TIC-1220)		55 °C	62 °C
Treated effluent pH (AIC-1224)	pH 6.5	pH 7.0	pH 8.5
Equalization tank level (LIC-1223)	20 %	52 %	85 %